

ELECTRONICS THAT CAN SURVIVE EXTREME TEMPERATURES

The benefits of extreme-temperature electronics are many—elimination of auxiliary cooling with massive heat-sink, lighter weight and smaller size, and integration of sensors and other transducers directly at the location of interest. Take a look at its applications and related developments

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Electronic devices when in use generate heat. These are normally equipped to dissipate this heat, but fail to do so completely. Unfortunately, circuits on a computer chip have temperature-dependent performance: when chips become too hot, they slow down. PCs are usually air-cooled, relying on simple fans to prevent the electronics from overheating, but for larger systems such as servers, server farms and data centres, forced air cooling is sometimes not sufficient. Air-cooling methods, which use fans to blow air across the chips, are not ideal for supercomputers or large server banks. Servers and data centres consume enormous amounts of power simply to run the electronics and keep them from overheating. Thus there is a need to develop extreme-temperature electronic (ETE) components for high-temperature applications.

Reducing the heat generated by CPUs of computers is an important research area now, and all the major chip makers are actively developing chips that consume less power and thus generate less heat. Designed to continuously operate for at least five years at 225°C, the standard electronic products family is targeted at sensor signal conditioning, data acquisition and control applications in hostile environments, and offers significant reliability and performance advantages over traditional silicon integrated circuits when the operating temperatures are greater than 150°C.



Nanotube memory chips (NRAMs) are impervious to the extreme environment and lethal hazards of space, as demonstrated by NASA during a recent shuttle mission

Although the present market for extreme-temperature electronics is small, it could develop tremendously if the automotive market opens up. The benefits of deploying extreme-temperature electronics, to say the least, include elimination of auxiliary cooling with massive heat-sink or heat-pipe design, lighter weight and smaller size, and integration of sensors and other transducers directly at the crucial location of interest.

Temperature limits of electronics

The term 'extreme-temperature electronics' refers to electronics operating outside the traditional temperature range of -55/-65°C to +125°C. It covers

very low temperatures, down to essentially absolute zero (0°K or -273°C), as well as high temperatures, from +125°C up to as high as electronics can be made to operate at.

At the low end, operation of semiconductor-based devices and circuits has often been reported down to temperatures as low as a few degrees above absolute zero (about -270°C). This includes devices based on Si, Ge, GaAs and other semiconductor materials.

Moreover, there is no reason to believe that operation should not extend all the way down to absolute zero.

At the high end, laboratory operation of discrete semiconductor devices has been reported at temperatures as high as +700°C (for a diamond Schottky diode) and 650°C (for a SiC MOSFET). Integrated circuits based on Si and GaAs have been reported to operate at +400 to +500°C. Si ICs, in particular, have been reported to operate at +300°C for 1000 hours or longer. Covering both extremes, there are reports of the same transistor working from about -270°C to +350 or even 400°C—an operating temperature span of over +600°C. Also, many passive components are useable to the lowest